

Laboratoire de Chimie Théorique

Relativistic electronic-structure theory including quantum electrodynamics corrections

Timothée Audinet

July 10, 2025

Thesis work with Julien Toulouse

1. Introduction
2. The 1D model: analytic results
3. The 1D model: numerical results
4. Vacuum-polarization density in 3D
5. Conclusion and Perspectives

1. Introduction

2. The 1D model: analytic results

3. The 1D model: numerical results

4. Vacuum-polarization density in 3D

5. Conclusion and Perspectives

Why are we interested in relativistic quantum chemistry?

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- Explains some physical properties of heavy elements

① yellow color of gold

P. Pyykkö, ACIE, (2004)

P. Romaniello and P. L. de Boeij, JCP, (2005)

② lead-acid battery electro-chemical potential

R. Ahuja, A. Blomqvist, P. Larsson, P. Pyykkö and P. Zaleski-Ejgierd, PRL, (2011)

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- Includes the electron spin and spin-orbit coupling

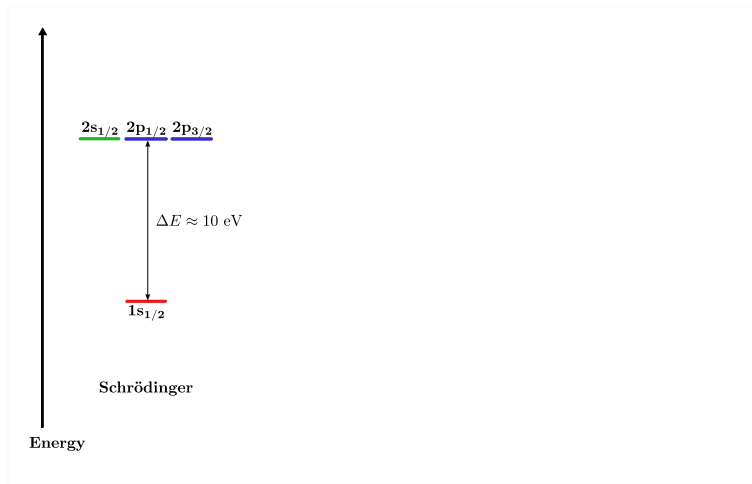


Figure 1: Lamb shift in hydrogen spectra

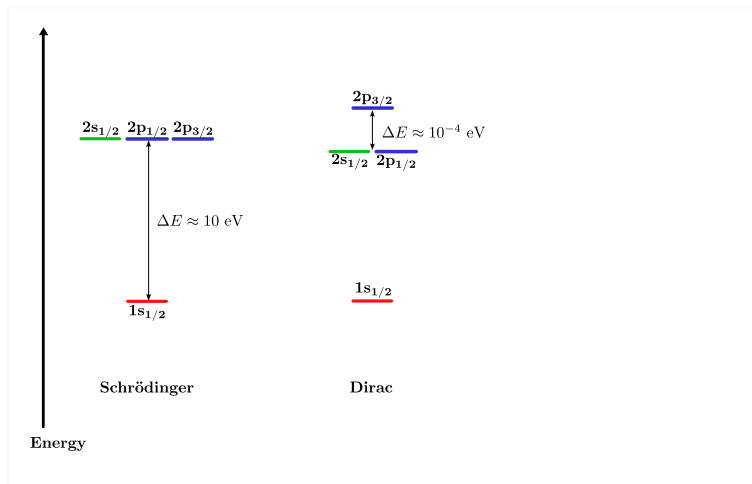


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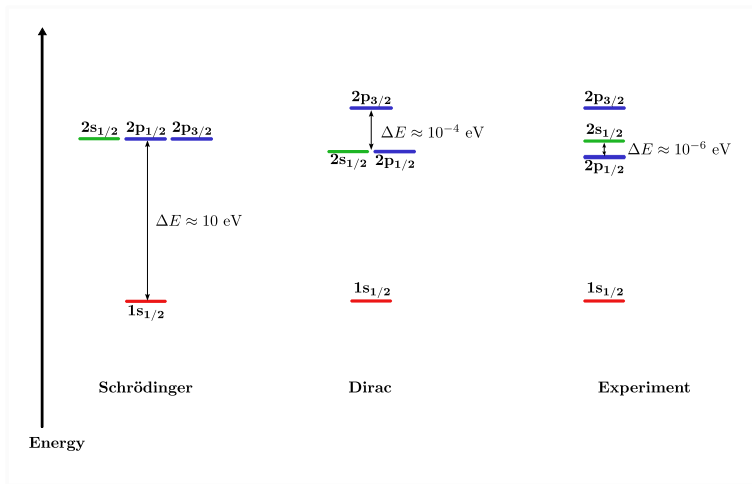


Figure 1: Lamb shift in hydrogen spectra

	IP	Error	EA	Error
DC-HF	7.6892	−1.5363	0.6690	−1.6396
DC-CCSD	9.1164	−0.1092	2.1070	−0.2017
DC-CCSD(T)	9.2938	0.0683	2.3457	0.0371
DC-CCSDTQP	9.2701	0.0446	2.3278	0.0192
+Breit	9.2546	0.0290	2.3188	0.0102
+QED	9.2288	0.0032	2.3072	−0.0014
Experiment [31,32]	9.2256		2.3086	

Figure 2: IP and EA of gold with different levels of approximation¹

¹P Schwerdtfeger *et al.*, PRL, (2017)

Non-relativistic spectrum

$$H_Z = \frac{\vec{p}^2}{2m} - \frac{Z}{|\mathbf{r}|} \quad (1)$$

Spectrum:

- If $\mathcal{E} > 0$: continuum
- If $\mathcal{E} < 0$: bound states

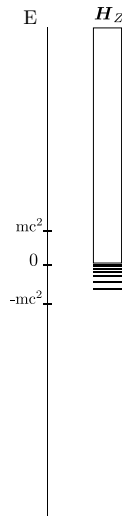
Relativistic spectrum

$$D_Z = c(\vec{\alpha} \cdot \vec{p}) + \beta mc^2 - \frac{Z}{|\mathbf{r}|} \mathbb{I}_4 \quad (2)$$

Spectrum:

- If $\mathcal{E} \in (-\infty, -mc^2] \cup [mc^2, +\infty)$: continuum
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⇒ Negative continuum spectrum



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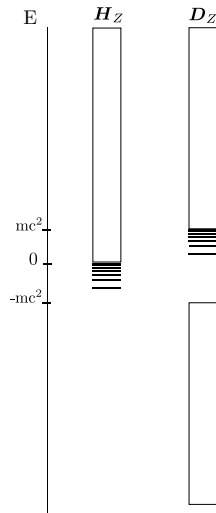
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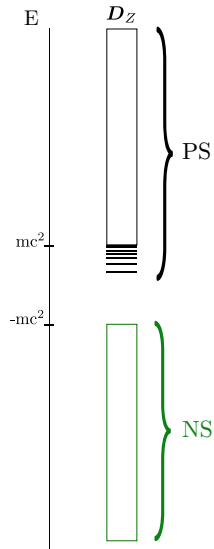
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One-electron Dirac operator

$$D_Z = c (\vec{\alpha} \cdot \vec{p}) + \beta mc^2 - \frac{Z}{|\mathbf{r}|} \mathbb{I}_4, \quad (3)$$

- $\vec{\alpha} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}$ where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ being the Pauli matrices
- $\beta = \begin{pmatrix} \mathbb{I}_2 & 0 \\ 0 & -\mathbb{I}_2 \end{pmatrix}$



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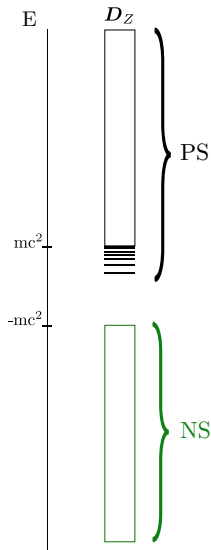
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One-electron Dirac equation

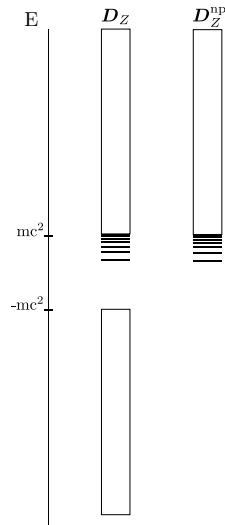
$$D_Z \psi = \mathcal{E} \psi \quad (4)$$

- $\psi = \begin{pmatrix} \psi^L \\ \psi^S \end{pmatrix}$ where $\psi^{L/S} = \begin{pmatrix} \psi_{\alpha}^{L/S} \\ \psi_{\beta}^{L/S} \end{pmatrix}$
- $\mathcal{E}$ is the energy of the state ψ
 - if $\mathcal{E} > 0$, $\psi \in \text{PS}$
 - if $\mathcal{E} < 0$, $\psi \in \text{NS}$



- No-pair field operator:

$$\hat{\psi}_{+}(\mathbf{r}) = \sum_{p \in \text{PS}} \hat{b}_p \psi_p(\mathbf{r}) \quad (5)$$



²Brown, Ravenhall, PRSL (1951)

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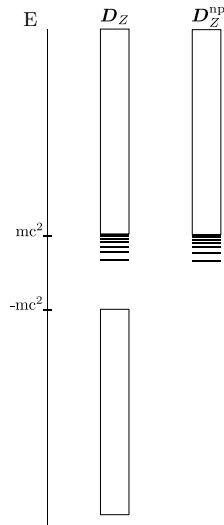
- No-pair many-electron Dirac-Coulomb-Breit Hamiltonian:

$$\hat{H}^{\text{np}} = \text{Tr}[\mathbf{D}_Z \hat{\mathbf{n}}_1^{\text{np}}] + \frac{1}{2} \text{Tr}[\mathbf{w}^{\text{CB}} \hat{\mathbf{n}}_2^{\text{np}}] \quad (6)$$

$$* \mathbf{w}^{\text{CB}}(\mathbf{r}_{12}) = \frac{1}{r_{12}} \mathbb{I}_4 \otimes \mathbb{I}_4 - \frac{1}{2r_{12}} \left(\vec{\alpha} \otimes \vec{\alpha} + \frac{(\vec{\alpha} \cdot \mathbf{r}_{12}) \otimes (\vec{\alpha} \cdot \mathbf{r}_{12})}{r_{12}^2} \right),$$

$$* \hat{\mathbf{n}}_1^{\text{np}} = \hat{\psi}_+^\dagger \otimes \hat{\psi}_+$$

$$* \hat{\mathbf{n}}_2^{\text{np}} = \hat{\psi}_+^\dagger \otimes \hat{\psi}_+^\dagger \otimes \hat{\psi}_+ \otimes \hat{\psi}_+$$



²Brown, Ravenhall, PRSL (1951)

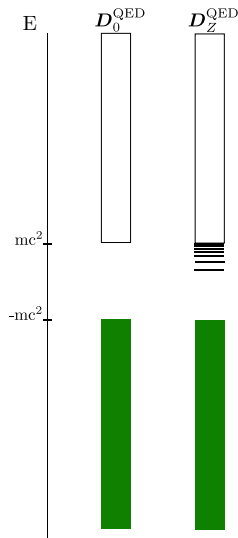
Free Dirac operator

$$D_0 = c (\vec{\alpha} \cdot \vec{p}) + \beta mc^2 \quad (7)$$

- generates the **free Dirac sea** once the negative states are filled: $|0\rangle$.
- QED field operator

$$\hat{\psi}(\mathbf{r}) = \sum_{p \in \text{PS}} \hat{b}_p \psi_p^0(\mathbf{r}) + \sum_{p \in \text{NS}} \hat{d}_p^\dagger \psi_p^0(\mathbf{r}) \quad (8)$$

- * $\{\hat{b}_p\}$ are the electronic operators, such that: $\hat{b}_p|0\rangle = 0$
- * $\{\hat{d}_p\}$ are the positronic operators, such that: $\hat{d}_p|0\rangle = 0$



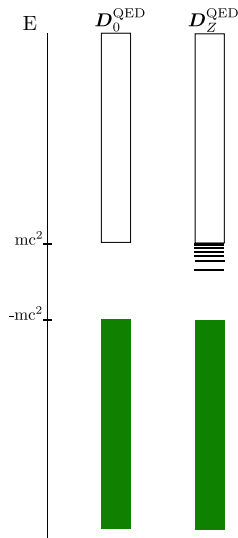
- Full QED picture without the photon degrees of freedom
- Effective QED Dirac-Coulomb-Breit Hamiltonian

$$\hat{H}^{\text{eQED}} = \text{Tr}[\mathbf{D}_Z \hat{\mathbf{n}}_1] + \frac{1}{2} \text{Tr}[\mathbf{w}^{\text{CB}} \hat{\mathbf{n}}_2] \quad (9)$$

$$* \hat{\mathbf{n}}_1 = \mathcal{N}_0[\hat{\psi}^\dagger \otimes \hat{\psi}] = \hat{\psi}^\dagger \otimes \hat{\psi} - \langle 0 | \hat{\psi}^\dagger \otimes \hat{\psi} | 0 \rangle$$

$$* \hat{\mathbf{n}}_2 = \mathcal{N}_0[\hat{\psi}^\dagger \otimes \hat{\psi}^\dagger \otimes \hat{\psi} \otimes \hat{\psi}]$$

- Normal-ordering $\mathcal{N}_0$ is taken w.r.t. $|0\rangle$ such that: $\langle 0 | \hat{H}^{\text{eQED}} | 0 \rangle = 0$.



³ Chaix, Iracane, JPB (1989) ; Saue, Visscher, INC (2003) ; Hainzl, Lewin, Seré, Solovej, PRA (2007) ; Liu, Lindgren, JCP (2013) ; Toulouse, SPC (2021)

- Full QED picture without the photon degrees of freedom
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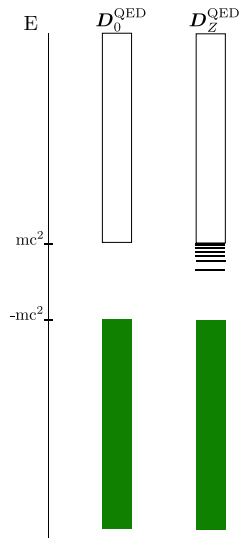
- Normal-ordering $\mathcal{N}_0$ is taken w.r.t. $|0\rangle$ such that: $\langle 0 | \hat{H}^{\text{eQED}} | 0 \rangle = 0$.

Total energy and minimization

Within the N -negatively charged Fock space sector $\mathcal{F}_N$, we have

$$E_N = \min_{\Psi \in \mathcal{F}_N; \langle \Psi | \Psi \rangle = 1} \langle \Psi | \hat{H}^{\text{eQED}} | \Psi \rangle \quad (10)$$

$\Rightarrow$ We can do quantum chemistry based on eQED



³Chaix, Iracane, JPB (1989) ; Saue, Visscher, INC (2003) ; Hainzl, Lewin, Seré, Solovej, PRA (2007) ; Liu, Lindgren, JCP (2013) ; Toulouse, SPC (2021)

- We need accurate QED calculations for high precision spectroscopy
- So far, state of the art QED for many-electrons systems is based on no-pair approximation
- **Next challenge in relativistic quantum chemistry:** can we go beyond the no-pair approximation?

Goals

- Understand the divergences appearing in eQED,
- Put relativistic quantum chemistry on a deeper theoretical ground,
- Develop ab initio methods based on eQED.

To do list:

- ☐ Study the 1D hydrogen-like system and compute the QED effects.
- ☐ Implement eQED in a finite basis set.
- ☐ Use our insight to expand the 3D vacuum-polarization density in a finite basis set.

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Free Dirac operator

$$D_{0,x} = c \boldsymbol{\alpha}_x p_x + \boldsymbol{\beta} mc^2, \quad (11)$$

- From this 1D operator we can make an unitary transformation

$$\mathbf{U} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

- Define a new operator $D'_{0,x} = \mathbf{U} D_{0,x} \mathbf{U}^{-1} = \begin{pmatrix} \mathbf{D}_0 & \mathbf{0}_2 \\ \mathbf{0}_2 & \mathbf{D}_0 \end{pmatrix}$

- Where

$$\mathbf{D}_0 = c \boldsymbol{\sigma}_x p_x + \boldsymbol{\sigma}_z mc^2$$

is a 2×2 operator!

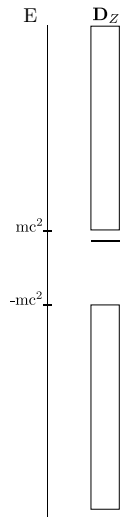
⁴Audinet, Toulouse, JCP (2023)

1D Dirac operator and eigenvalue

$$\mathbf{D}_Z = -ic\sigma_x \frac{d}{dx} + \sigma_z mc^2 - Z\delta(x) \quad (12)$$

$$\mathbf{D}_Z \psi_p^Z = \varepsilon_p \psi_p^Z \quad (13)$$

- Bound State: $\varepsilon_b \in (-mc^2, mc^2)$
- Continuum: $\varepsilon_k = \pm \sqrt{m^2 c^4 + k^2 c^2}$



⁵Lapidus, AJP (1983)

1D Dirac operator and eigenvalue

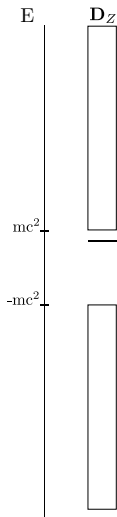
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The delta-type interaction

- In the non-relativistic limit, recovers the energy and radial part of wave-function, including the cusp, of the ground-state of 3D hydrogen-like system
Herrick, Stillingner, PRA, (1975)
Traore, Giner, Toulouse, JCP, (2022)
- Some analytical results are known
Nogami, Beachey, EL, (1986)



⁵Lapidus, AJP, (1983)

Second quantized Hamiltonian

$$\hat{H} = \text{Tr}[\mathbf{D}_Z \hat{\mathbf{n}}_1] + \frac{1}{2} \text{Tr}[\mathbf{w} \hat{\mathbf{n}}_2] \quad (14)$$

with the normal-ordered 1D density operators:

- $\hat{\mathbf{n}}_1 = \mathcal{N}_0[\hat{\psi}^\dagger \otimes \hat{\psi}]$
- $\hat{\mathbf{n}}_2 = \mathcal{N}_0[\hat{\psi}^\dagger \otimes \hat{\psi}^\dagger \otimes \hat{\psi} \otimes \hat{\psi}]$

and the 1D two-particle Coulomb-Breit interaction:

- $\mathbf{w}(x_{12}) = \delta(x_{12})(\mathbf{I}_2 \otimes \mathbf{I}_2 - \sigma_x \otimes \sigma_x)$

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Hartree-Fock energy

$$E_N^{\text{HF}} = \min_{\Phi \in \mathcal{S}_N} \langle \Phi | \hat{H} | \Phi \rangle = \text{Tr}[\mathbf{D}_Z \mathbf{n}_1^{\text{HF}}] + \frac{1}{2} \text{Tr}[\mathbf{w} \mathbf{n}_2^{\text{HF}}]$$

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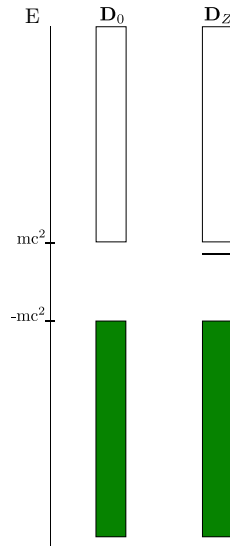
Perturbation theory (on $\mathbf{w}$)

$$\mathcal{E}_N^{(1)} = E_N^{(1)} - E_0^{(1)}$$

Density matrix of a N -negatively charged system

$$\mathbf{n}_1(x, x') = \mathbf{n}_1^{\text{el}}(x, x') + \mathbf{n}_1^{\text{vp}}(x, x') \quad (15)$$

- $\mathbf{n}_1^{\text{el}}(x, x') = \sum_{i=1}^N \psi_i^Z(x) \psi_i^{Z\dagger}(x')$
- $\mathbf{n}_1^{\text{vp}}(x, x') = \sum_{p \in \text{NS}} \psi_p^Z(x) \psi_p^{Z\dagger}(x') - \sum_{p \in \text{NS}} \psi_p^0(x) \psi_p^{0\dagger}(x')$



Density matrix of a N -negatively charged system

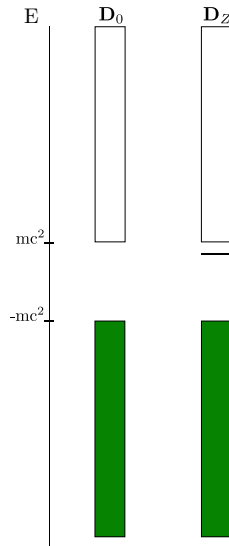
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Relative energies

$$\mathcal{E}_N^{(0)} = \text{Tr}[\mathbf{D}_Z \mathbf{n}_1^{\text{el}}] \quad (16)$$

$$\mathcal{E}_N^{(1)} = \frac{1}{2} \text{Tr}[\mathbf{w} \mathbf{n}_2^{\text{el}}] + \text{Tr}[\mathbf{w} \mathbf{n}_2^{\text{el/vp}}] \quad (17)$$

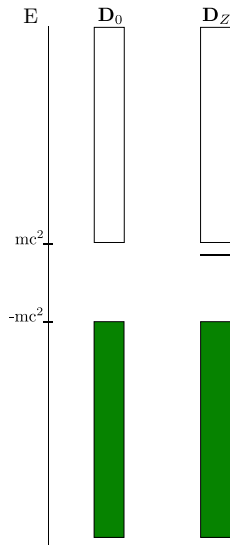


- Spontaneous creation of electron positron pairs due to the external potential

Vacuum-polarization density matrix

$$\mathbf{n}_1^{\text{vp}}(x, x') = \sum_{p \in \text{NS}} \psi_p^Z(x) \psi_p^{Z\dagger}(x') - \sum_{p \in \text{NS}} \psi_p^0(x) \psi_p^{0\dagger}(x')$$

- Creates a charge density that will interact through the two-particle interaction with the bound electrons of the system
- Responsible for energy shifts in the electronic structure of atoms and molecules



First-order vacuum-polarization correction

$$\mathcal{E}_N^{\text{vp},(1)} = \text{Tr} \left[\mathbf{w}(x_{12}) \mathbf{n}_2^{\text{el/vp}}(x_1, x_2) \right] \quad (18)$$

where

- $\mathbf{n}_2^{\text{el/vp}} = \mathbf{n}_1^{\text{el}} \otimes \mathbf{n}_1^{\text{vp}} - \mathbf{X}(\mathbf{n}_1^{\text{el}} \otimes \mathbf{n}_1^{\text{vp}})$
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Direct terms

$$\mathcal{E}_N^{\text{vp},(1),\text{DC}} = \int n^{\text{el}}(x) n^{\text{vp}}(x) dx \quad (19)$$

$$\mathcal{E}_N^{\text{vp},(1),\text{DB}} = -\frac{1}{c^2} \int j^{\text{el}}(x) j^{\text{vp}}(x) dx \quad (20)$$

Exchange terms

$$\mathcal{E}_N^{\text{vp},(1),\text{XC}} = - \int \text{tr}[\mathbf{n}_1^{\text{el}}(x) \mathbf{n}_1^{\text{vp}}(x)] dx \quad (21)$$

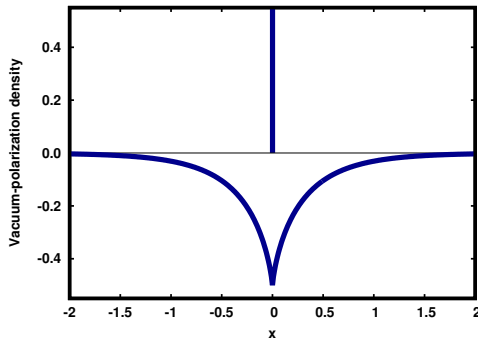
$$\mathcal{E}_N^{\text{vp},(1),\text{XB}} = \frac{1}{c^2} \int \text{tr}[\mathbf{j}_1^{\text{el}}(x) \mathbf{j}_1^{\text{vp}}(x)] dx \quad (22)$$

- where $\mathbf{j}_1 = c\sigma_x \mathbf{n}_1$ is the current-density matrix

First-order perturbation theory w.r.t Z of the vacuum-polarization density

$$n^{\text{vp},(1)}(x) = \sum_{p \in \text{NS}} \psi_p^{Z,(1)\dagger}(x) \psi_p^{Z,(1)}(x) - \sum_{p \in \text{NS}} \psi_p^{0\dagger}(x) \psi_p^0(x) = \frac{Z}{\pi c} \delta(x) + n_{\text{reg}}^{\text{vp},(1)}(x) \quad (23)$$

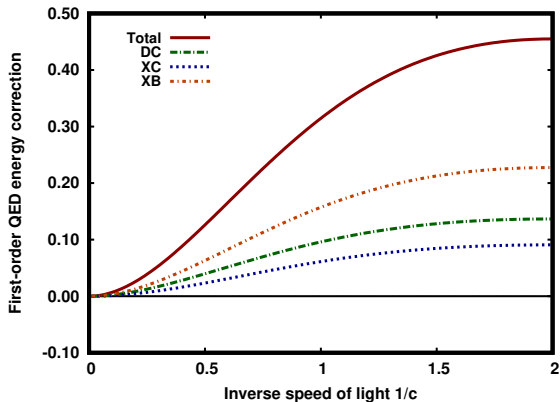
Audinet, Morellini, Levitt, Toulouse, JPA (2025)



Lamb shift-type energy shift

Correction to the ground-state energy due to the interaction between the electrons and the vacuum-polarization density

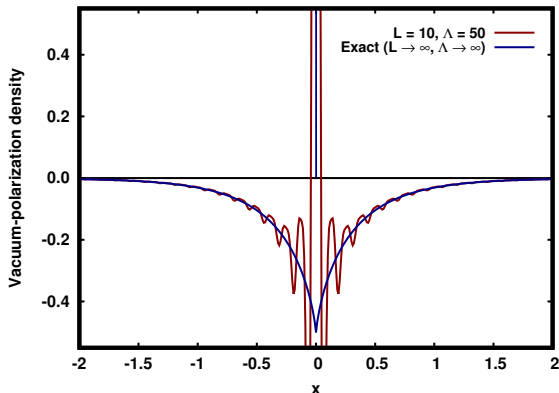
$$\mathcal{E}_N^{\text{vp},(1)} = \text{Tr}[\mathbf{w} \mathbf{n}_2^{\text{el/vp}}] \quad (24)$$



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Plane-waves basis

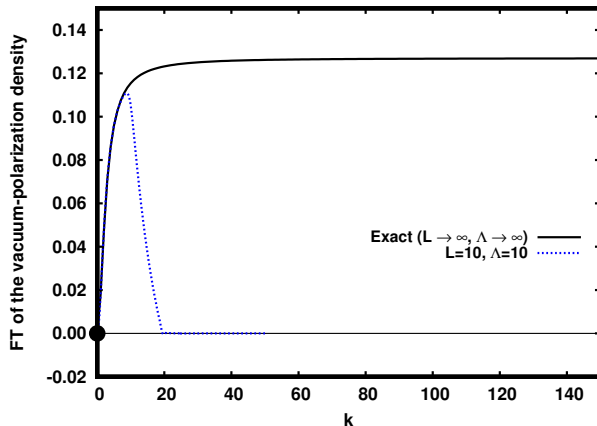
$$\forall x \in \mathbb{R}, k \in \frac{2\mathbb{Z}\pi}{L}, |k| \leq \Lambda, \chi_k(x) = \frac{1}{\sqrt{L}} e^{ikx} \quad (25)$$



Regularization with Fourier Transform

We regularize the vacuum-polarization density by removing the large-momentum contributions

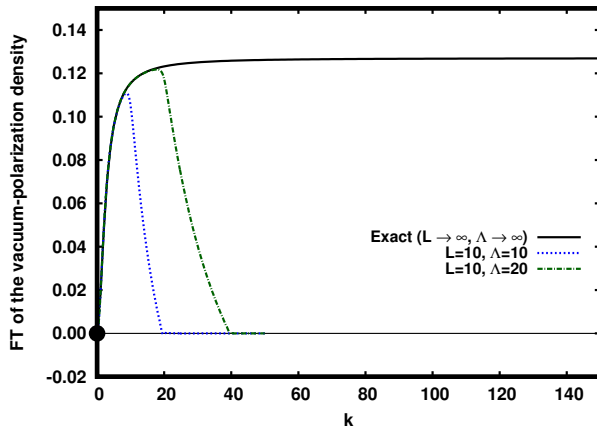
$$\hat{n}_{\text{reg}}^{\text{vp}}(k) = \left[\hat{n}^{\text{vp}}(k) - \hat{n}^{\text{vp}}(k_{\text{max}}) \right] \theta(k_{\text{max}} - k) \quad (26)$$



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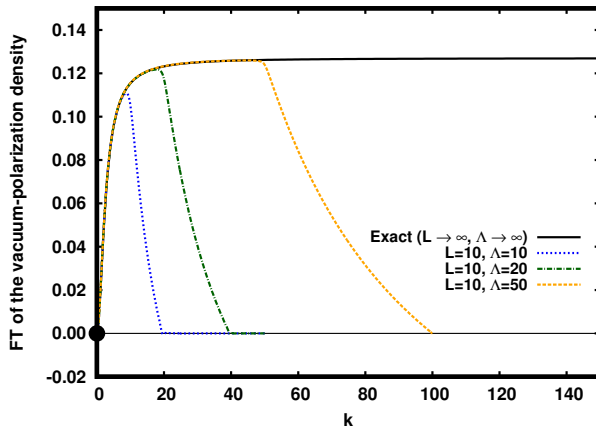
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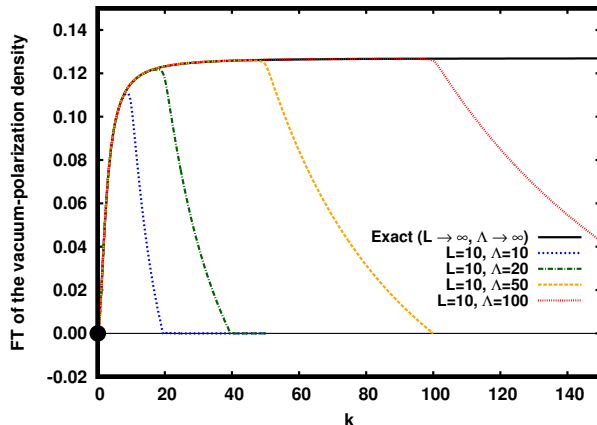
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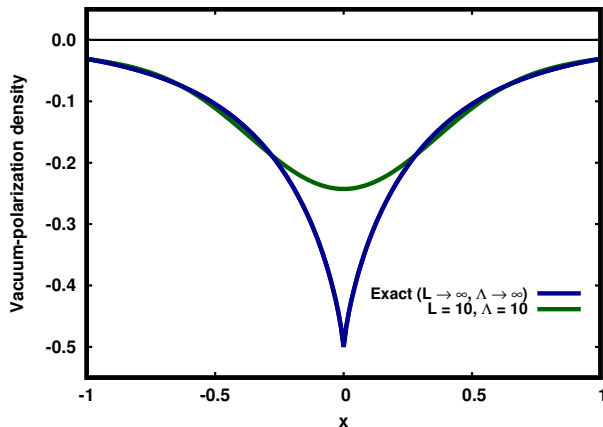
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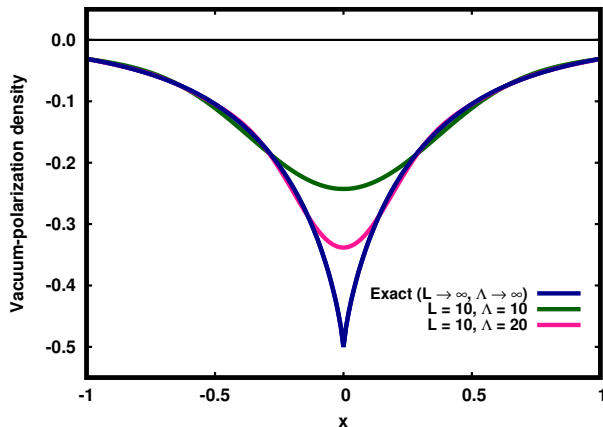
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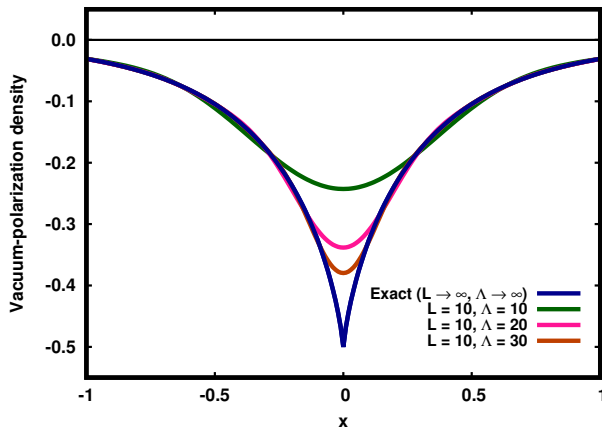
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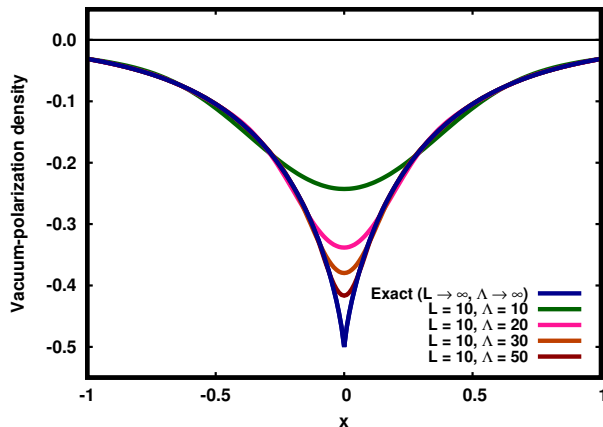
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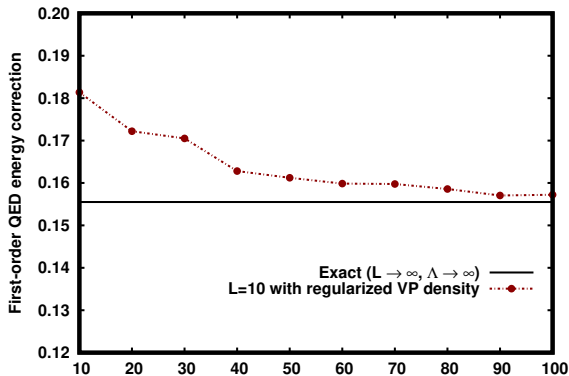
$$n_{\text{reg}}^{\text{vp}}(x) = \mathcal{F}^{-1} [\hat{n}_{\text{reg}}^{\text{vp}}(k)] \quad (27)$$



Regularization of the vacuum-polarization density matrix

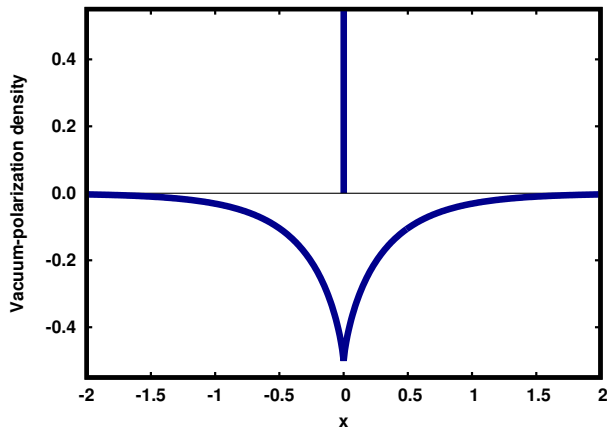
Using the regularized vacuum-polarization density matrix in Fourier space

$$\hat{n}_{\text{reg},L,\Lambda}^{\text{vp}}(k) = \left[\hat{n}_{L,\Lambda}^{\text{vp}}(k) - \hat{n}_{L,\Lambda}^{\text{vp}}(k_{\text{max}}) \right] \theta(k_{\text{max}} - |k|) \quad (28)$$



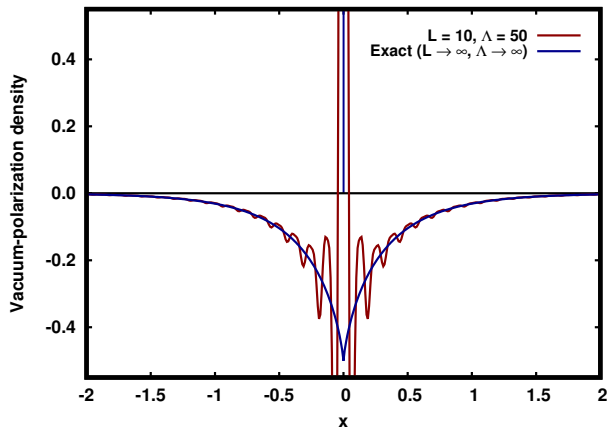
Results

- Found analytical solutions to the 1D model
- Manage to converge QED effects in a finite basis



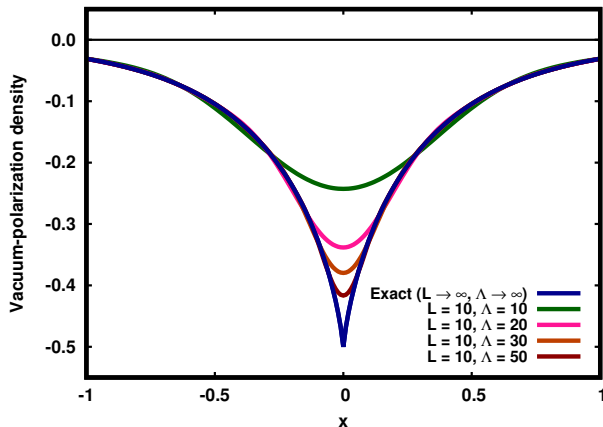
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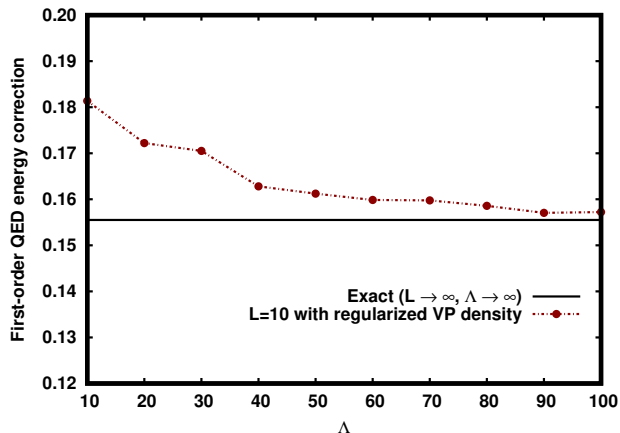
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1. Introduction
2. The 1D model: analytic results
3. The 1D model: numerical results
- 4. Vacuum-polarization density in 3D**
5. Conclusion and Perspectives

Vacuum polarization in Hydrogen-like atoms in 3D

- In 3D the vacuum-polarization density matrix diverges
- In practice it is expanded as a perturbative expansion over the nuclear charge Z

$$n^{\text{vp}}(\mathbf{r}) = n^{\text{vp},(1)}(\mathbf{r}) + n^{\text{vp,WK}}(\mathbf{r}) \quad (29)$$

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- Contains all the divergences
- Need to be regularized and renormalized

$$n^{\text{vp},(1)} \approx n^{\text{vp,U}} + \text{div.} \quad (30)$$

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Ultraviolet renormalization

- Using a UV cutoff in Fourier space

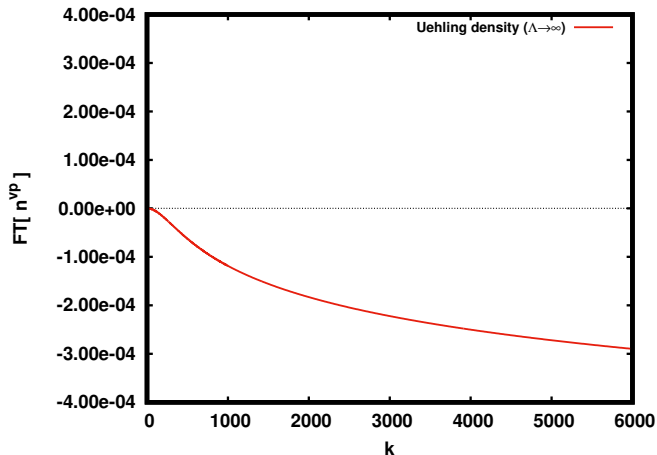
$$n^{\text{vp},(1)}(\mathbf{r}) \rightarrow \hat{n}_{\Lambda}^{\text{vp},(1)}(\mathbf{k}) \quad (31)$$

- To identify the singularities
- Simulate the finiteness of a basis set

⁶Uehling, PR, (1935)

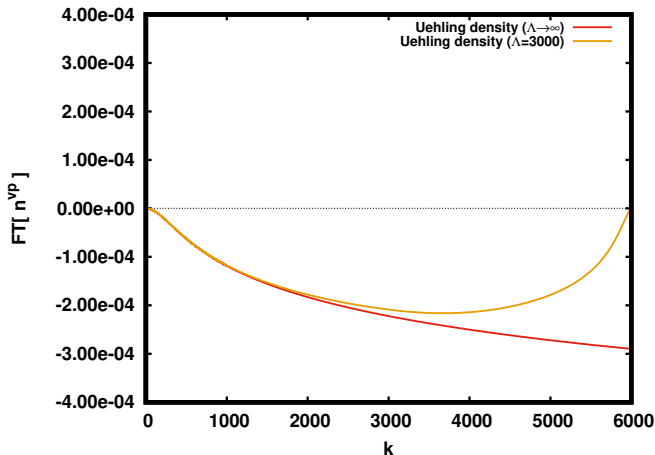
The truncated first-order vacuum-polarization density

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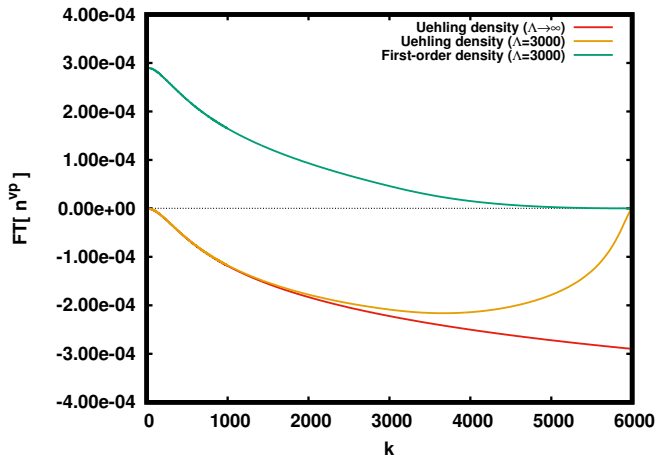
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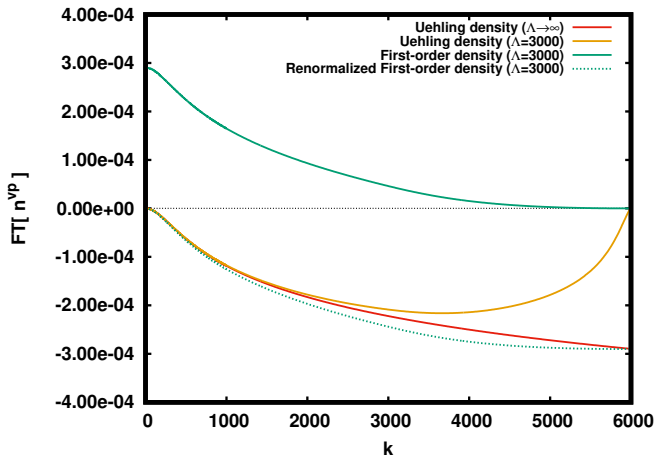
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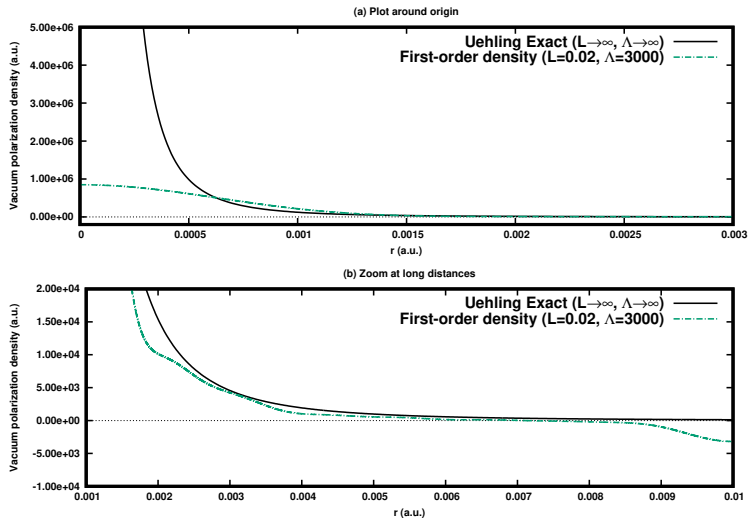
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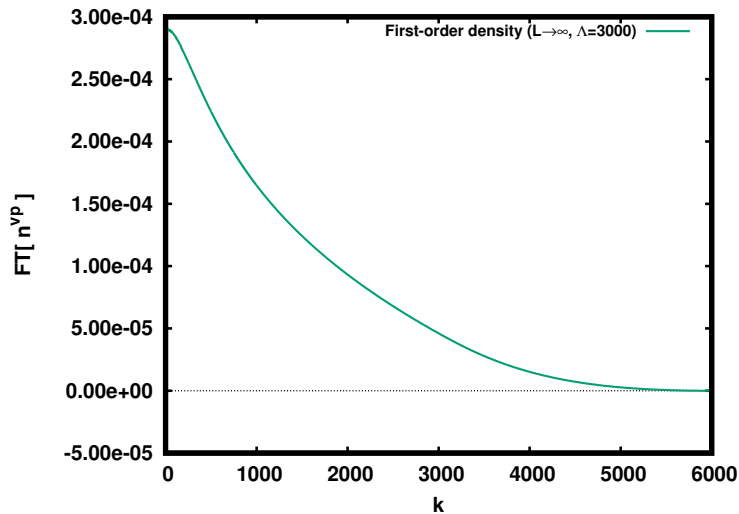


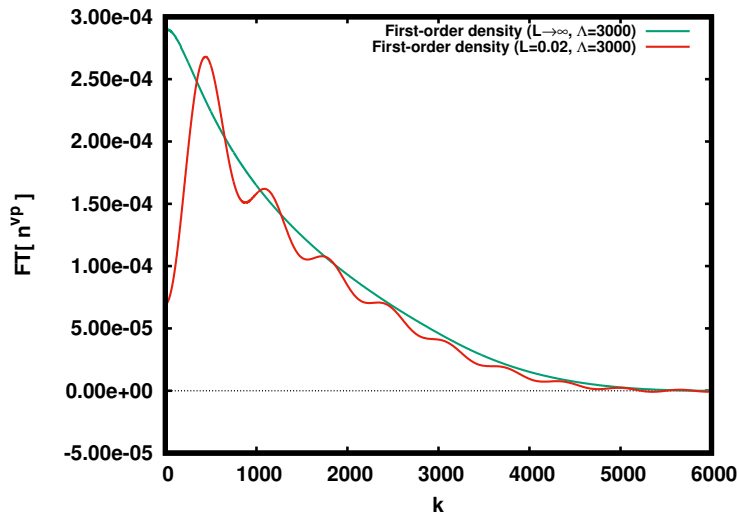
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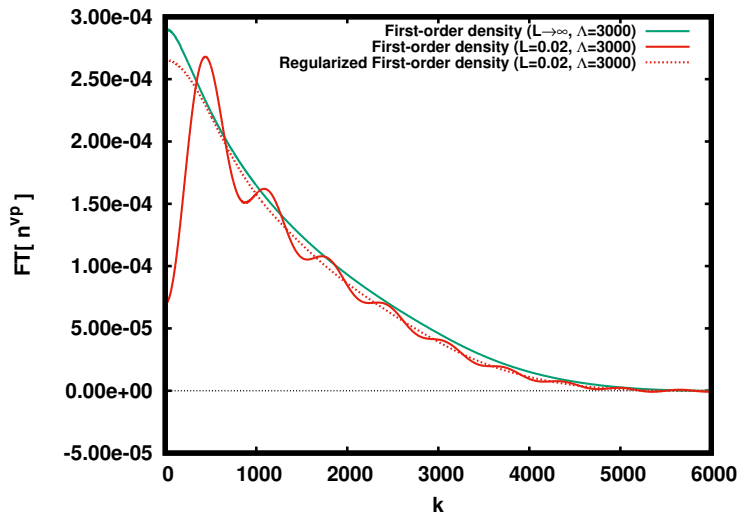
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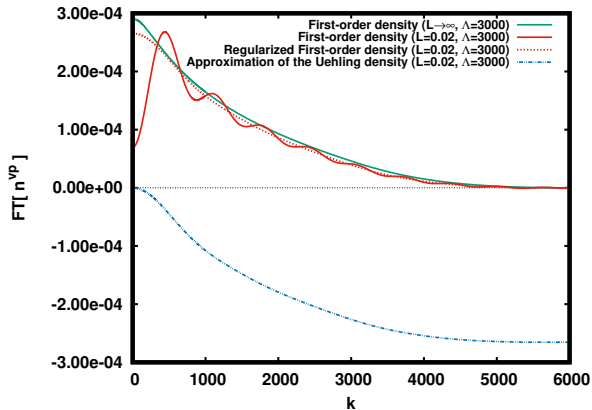


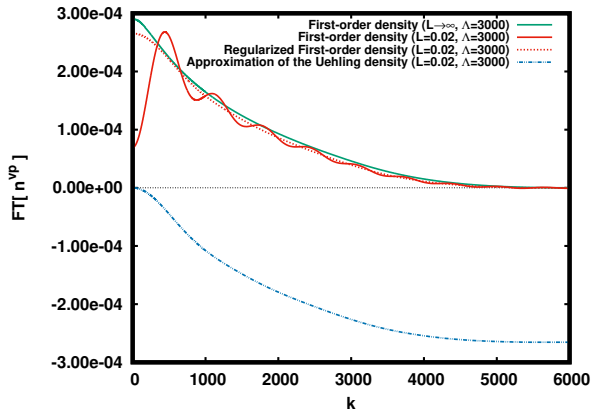












UV cutoff (Λ)	$\mathcal{E}_{1,1s}^{vp,(1)} (10^{-8} \text{ a.u.})$
1000	-1.54
1500	-1.51
2000	-1.49
2500	-1.52
3000	-1.40
∞	-3.27

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To do list

- ✓ Study the 1D hydrogen-like system and compute the QED effects.
- ✓ Implement eQED in a finite basis set.
- Use our insight to expand the 3D vacuum-polarization density in a finite basis set (in progress).

- We developed an effective QED theory including electron-positron pairs.
- Some singularities appears in a finite basis.
- This 1D model help us to understand this problem.

- Reaching convergence of the energy shift in a finite basis set
 - * Either by increasing the maximum cutoff
 - * Improving the regularization procedure
 - * Using different basis set (Gaussian types)

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- Generalization of the code to include electronic correlation for atoms and molecules in a fully eQED description

Thank you for your attention!